

On a Subset-Sum Reduction for Equilibrium Existence in Fisher Markets with Utility and Earning Limits

Verification and repaired write-up of the zero-budget construction

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Abstract

Garg, Hoefer, and Mehlhorn ask whether, beyond money-clearing markets, deciding existence of a thrifty and modest equilibrium in a linear Fisher market with both utility limits and earning limits is NP-hard or instead admits a simple polynomial-time-checkable characterization. This note verifies a proposed reduction from SUBSET SUM. The reduction is correct, as a weak NP-hardness proof, for the model in which buyer budgets are allowed to be zero. In the stricter convention in which every buyer must have strictly positive budget, the reduction is not yet a proof: its zero-price blockers have budget zero. We isolate the precise gap, give a clean complete proof for the nonnegative-budget model, and record a rigorous positive-budget blocker lemma together with the obstruction that prevents the standard “small perturbation” sentence from being a complete repair. Thus the result below is a solution for the nonnegative-budget variant and partial progress, not a complete solution for the strictly-positive-budget formulation.

1 Model

A market consists of a finite set B of buyers and a finite set G of divisible goods, each good having unit supply. Buyer i has a budget $m_i \geq 0$, a utility cap $c_i > 0$, and values $u_{ij} \geq 0$. Good j has earning cap $d_j > 0$. For an allocation $x = (x_{ij})$, buyer i ’s utility is

$$U_i(x_i) = \min \left\{ c_i, \sum_{j \in G} u_{ij} x_{ij} \right\}.$$

At prices $p \geq 0$, a demand bundle of buyer i is an affordable bundle maximizing U_i . A demand bundle is *modest* if the uncapped linear utility in the displayed sum is at most c_i , and it is *thrifty* if it uses only maximum-bang-per-buck goods. For a seller, a modest supply at price p_j is

$$e_j = \min\{1, d_j/p_j\},$$

with the usual convention that $e_j = 1$ when $p_j = 0$. A thrifty and modest equilibrium is a pair (x, p) such that prices are nonnegative, every buyer receives a thrifty and modest demand bundle, every seller uses modest supply, no good is overallocated, and Walras’ law

$$p_j(e_j - \sum_i x_{ij}) = 0$$

holds for every good j .

The source paper’s displayed model uses $m_i > 0$ in its formal description. The proof below deliberately proves the theorem for the slightly relaxed convention $m_i \geq 0$. This distinction is the main status issue discussed in Section 6.

2 Active-money form at positive prices

For positive prices, define money flows $f_{ij} = p_j x_{ij}$ and inverse MBB variables

$$\lambda_i = \min_{j: u_{ij} > 0} \frac{p_j}{u_{ij}}.$$

The active budget of buyer i and the active earning of good j are

$$a_i(\lambda_i) = \min\{m_i, c_i \lambda_i\}, \quad b_j(p_j) = \min\{d_j, p_j\}.$$

Lemma 1 (active-money characterization). *Suppose all goods with positive value to some buyer have positive price. Then (x, p) is a thrifty and modest equilibrium if and only if there are nonnegative flows f_{ij} such that*

$$p_j \geq u_{ij} \lambda_i \quad \text{for all } i, j, \quad (1)$$

$$f_{ij} > 0 \implies p_j = u_{ij} \lambda_i, \quad (2)$$

$$\sum_j f_{ij} = \min\{m_i, c_i \lambda_i\} \quad \text{for every buyer } i, \quad (3)$$

$$\sum_i f_{ij} = \min\{d_j, p_j\} \quad \text{for every good } j. \quad (4)$$

The allocation is recovered by $x_{ij} = f_{ij}/p_j$.

Proof. At positive prices, spending one unit of money on an MBB good of buyer i gives utility $1/\lambda_i$. Hence a thrifty modest demand bundle spends exactly $\min\{m_i, c_i \lambda_i\}$: an uncapped buyer spends all its budget, while a capped buyer spends the minimum money needed to reach its cap. This gives (3), and thriftiness is exactly (1)–(2). For a positive-price good, Walras' law implies that the sold amount equals the seller's modest supply, so the money earned is $p_j \min\{1, d_j/p_j\} = \min\{d_j, p_j\}$, which is (4). Conversely, if flows satisfying (1)–(4) are given and $x_{ij} = f_{ij}/p_j$, then every buyer receives a thrifty modest demand bundle and every positive-price good is sold exactly at its modest supply. Walras' law follows. $\square$

3 The reduction

We reduce from SUBSET SUM. The input is a list of positive integers

$$a_1, \dots, a_n, T \quad \text{with} \quad 0 < T < \sum_{i=1}^n a_i.$$

Let

$$S = \sum_{i=1}^n a_i, \quad K = 2S + 1.$$

Then $K > T$ and $K > 2a_i$ for every i .

3.1 Goods

There is one public good g_0 , two private item goods A_i, B_i for each item i , and two private floor goods r, s . Their earning caps are

$$d_{g_0} = T, \quad d_{A_i} = K, \quad d_{B_i} = 2K - a_i, \quad d_r = 1, \quad d_s = 3.$$

All unspecified values are zero.

3.2 The floor buyer

There is a floor buyer F with

$$m_F = 2, \quad c_F = 2,$$

and values

$$u_{F,r} = 4, \quad u_{F,s} = 1, \quad u_{F,g_0} = 2K, \quad u_{F,A_i} = 2K, \quad u_{F,B_i} = K.$$

3.3 The item buyers

For each item i there is a buyer I_i with

$$m_i = 3K, \quad c_i = 2K,$$

and values

$$u_{I_i,A_i} = 2K, \quad u_{I_i,B_i} = K, \quad u_{I_i,g_0} = K.$$

Buyer I_i has zero value for the private goods of every other item.

3.4 Zero-price blockers

For every good $g \in \{g_0, r, s, A_1, B_1, \dots, A_n, B_n\}$ add two blocker buyers $Z_{g,1}, Z_{g,2}$ with

$$m_{Z_{g,h}} = 0, \quad c_{Z_{g,h}} = 1, \quad u_{Z_{g,h},g} = 1,$$

and zero value for all other goods.

Lemma 2 (all prices are positive). *In every thrifty and modest equilibrium of the constructed market, every good has positive price.*

Proof. If $p_g = 0$, then each of the two blockers $Z_{g,1}, Z_{g,2}$ can obtain its utility cap 1 at zero cost only by taking the full unit of good g . Since each blocker values only g , every modest demand bundle of each blocker contains one unit of g . The two demands cannot both be met from a unit-supply good. Thus $p_g > 0$ in equilibrium. If $p_g > 0$, a blocker with budget zero has active budget zero and receives the zero bundle. $\square$

4 Analysis of every equilibrium

By Lemma 2, the active-money characterization applies.

Lemma 3 (the floor buyer pins the scale). *In every equilibrium, $\lambda_F = 1$. The floor buyer spends one unit of money on r , one unit on s , and no money on any other good.*

Proof. The only non-blocker buyer who values r and s is F , and blockers spend zero at positive prices. Hence all active earning of r and s must be paid by F . If $\lambda = \lambda_F$, positive flow from F to r and s implies

$$p_r = 4\lambda, \quad p_s = \lambda.$$

Thus the mandatory spending of F on r and s is

$$G_F(\lambda) = \min\{1, 4\lambda\} + \min\{3, \lambda\},$$

whereas its active budget is

$$A_F(\lambda) = \min\{2, 2\lambda\}.$$

For $0 < \lambda < 1/4$, $G_F(\lambda) = 5\lambda > 2\lambda = A_F(\lambda)$. For $1/4 \leq \lambda < 1$, $G_F(\lambda) = 1 + \lambda > 2\lambda = A_F(\lambda)$. At $\lambda = 1$, $G_F(1) = A_F(1) = 2$. For $1 < \lambda < 3$, $G_F(\lambda) = 1 + \lambda > 2 = A_F(\lambda)$, and for $\lambda \geq 3$, $G_F(\lambda) = 4 > 2 = A_F(\lambda)$. Therefore the mandatory floor-good spending is affordable only when $\lambda = 1$. At $\lambda = 1$, the two floor goods alone exhaust F 's active budget. $\square$

Lemma 4 (item-buyer alternatives). *In every equilibrium, each item buyer I_i is in exactly one of the following two states.*

- (a) Low state: $\lambda_i = 1$, buyer I_i spends only on A_i and B_i , and it spends no money on g_0 .
- (b) High state: $\lambda_i = p_{g_0}/K \geq 2$, buyer I_i spends on A_i , B_i , and exactly a_i units of money on g_0 .

Proof. Fix i . By Lemma 3, buyer F has no active budget left after buying r and s . Blockers have zero budget. Therefore all active earning of A_i and B_i must be paid by I_i . Since these active earnings are positive, both goods are MBB for I_i , so

$$p_{A_i} = 2K\lambda_i, \quad p_{B_i} = K\lambda_i.$$

The MBB inequalities for the floor buyer, whose inverse MBB value is 1, give

$$p_{A_i} \geq 2K, \quad p_{B_i} \geq K,$$

and hence $\lambda_i \geq 1$.

The private spending forced on I_i is

$$G_i(\lambda_i) = \min\{K, 2K\lambda_i\} + \min\{2K - a_i, K\lambda_i\} = K + \min\{2K - a_i, K\lambda_i\},$$

because $\lambda_i \geq 1$. Its active budget is

$$A_i(\lambda_i) = \min\{3K, 2K\lambda_i\}.$$

The breakpoint $(2K - a_i)/K = 2 - a_i/K$ lies strictly between $3/2$ and 2 , since $K > 2a_i$.

Suppose first that I_i spends no money on g_0 . Then $G_i(\lambda_i) = A_i(\lambda_i)$. For $1 \leq \lambda_i \leq 3/2$, this equality is

$$K + K\lambda_i = 2K\lambda_i,$$

so $\lambda_i = 1$. For $3/2 < \lambda_i < 2 - a_i/K$, the equality is

$$K + K\lambda_i = 3K,$$

whose only solution is $\lambda_i = 2$, outside the interval. For $\lambda_i \geq 2 - a_i/K$, the equality is

$$3K - a_i = 3K,$$

which is impossible. Thus a no-public-spending item buyer is exactly a low-state buyer.

Now suppose I_i spends positive money on g_0 . Then g_0 is MBB for I_i , so

$$p_{g_0} = K\lambda_i.$$

The MBB inequality for F on g_0 gives $p_{g_0} \geq 2K$, and hence $\lambda_i \geq 2$. For every $\lambda_i \geq 2$,

$$A_i(\lambda_i) = 3K, \quad G_i(\lambda_i) = K + (2K - a_i) = 3K - a_i.$$

Therefore the missing active budget is exactly a_i , and this is precisely the amount that I_i must spend on g_0 . $\square$

5 Correctness

Theorem 1. *For linear Fisher markets with utility limits and earning limits in which buyer budgets are allowed to be zero, deciding existence of a thrifty and modest equilibrium is weakly NP-hard.*

Proof. We show that the constructed market has an equilibrium if and only if the SUBSET SUM instance has a subset $H \subseteq \{1, \dots, n\}$ with $\sum_{i \in H} a_i = T$.

Suppose first that an equilibrium exists. By Lemma 3, the public good satisfies

$$p_{g_0} \geq 2K > T = d_{g_0}.$$

Thus g_0 is earning-capped and receives exactly T units of money. The floor buyer spends no money on g_0 , and blockers have zero budget. Hence all T units of public-good spending come from item buyers. By Lemma 4, an item buyer spends money on g_0 exactly in the high state, and high-state buyer I_i spends exactly a_i on g_0 . Therefore

$$H = \{i : I_i \text{ is in the high state}\}$$

satisfies

$$\sum_{i \in H} a_i = T.$$

Conversely, suppose $H \subseteq \{1, \dots, n\}$ satisfies $\sum_{i \in H} a_i = T$. Set

$$\lambda_F = 1, \quad \lambda_i = 2 \text{ for } i \in H, \quad \lambda_i = 1 \text{ for } i \notin H.$$

Set prices

$$p_r = 4, \quad p_s = 1, \quad p_{g_0} = 2K,$$

and, for every item i ,

$$p_{A_i} = 2K\lambda_i, \quad p_{B_i} = K\lambda_i.$$

The floor buyer spends one unit on r and one unit on s . If $i \notin H$, buyer I_i spends K on A_i and K on B_i . If $i \in H$, buyer I_i spends K on A_i , $2K - a_i$ on B_i , and a_i on g_0 . The public-good spending is feasible because $\sum_{i \in H} a_i = T$.

Every good receives exactly its active earning: r earns 1, s earns 1, g_0 earns T , every A_i earns K , every low-state B_i earns $K = p_{B_i}$, and every high-state B_i earns $2K - a_i$. Every positive-budget buyer spends exactly its active budget: F spends 2, every low-state item buyer spends $2K$, and every high-state item buyer spends $3K$. All spending is on MBB goods. Blocker buyers have active budget zero and receive the zero bundle. Therefore the active-money conditions of Lemma 1 hold, so the corresponding allocation is a thrifty and modest equilibrium.

The construction has polynomial size and all displayed integers have polynomial bit length. This proves weak NP-hardness. $\square$

Corollary 1. *Unless $P = NP$, no necessary-and-sufficient characterization of equilibrium existence for the nonnegative-budget model can both apply to all instances and be checked in polynomial time.*

Proof. Such a characterization would decide the NP-hard language in polynomial time. $\square$

6 Status of the strictly-positive-budget version

The proof above uses zero budgets only for the blockers. In the source paper's formal model, however, the preliminary definition says that every buyer has budget $m_i > 0$. Therefore Theorem 1 should not be quoted as a complete solution for that stricter formulation.

A natural first repair is to replace each zero-budget blocker by a very small positive-budget blocker. The following lemma records the useful part of that idea.

Lemma 5 (positive-budget blocker accounting). *Fix a good g . Add two buyers, each with budget $\varepsilon > 0$, utility cap 1, value 1 for g , and value zero for all other goods. In any equilibrium in which g has earning cap at least 2ε , the price of g is at least 2ε , and the two blockers together spend exactly 2ε on g .*

Proof. If $p_g = 0$, both blockers demand the full unit of g , which is impossible. If $0 < p_g < \varepsilon$, each blocker is capped and demands one full unit; again impossible. If $\varepsilon \leq p_g < 2\varepsilon$, each blocker spends its whole budget ε , so the two blockers alone spend 2ε , while the active earning of the good is at most $p_g < 2\varepsilon$, impossible. Hence $p_g \geq 2\varepsilon$, and then each blocker is budget-limited and spends exactly ε . $\square$

This lemma is not by itself a repair of the reduction. If the earning cap of each watched good is increased by 2ε , then the blockers contribute a fixed 2ε at positive prices, and the residual earning left for the original buyers becomes

$$\min\{d_g, p_g - 2\varepsilon\},$$

not $\min\{d_g, p_g\}$. This additive shift changes the algebra of the floor module and creates additional low-scale solutions.

Proposition 1 (why the naive perturbation is incomplete). *Consider only the original floor buyer and the two floor goods r, s , and replace the zero-budget blockers on r and s by the positive-budget blockers of Lemma 5. Increase the earning caps of r and s by 2ε . For every $0 < 2\varepsilon < 1$, the floor submarket has a positive-price solution with*

$$\lambda_F = \varepsilon, \quad p_r = 4\varepsilon, \quad p_s = 2\varepsilon.$$

In this solution, s is filled entirely by its two blockers and the floor buyer spends its whole active budget 2ε on r . Thus the original conclusion $\lambda_F = 1$ is no longer forced.

Proof. Let $W = 2\varepsilon$. At the proposed prices, the two blockers on each good spend W in total. The residual active earning of r after blocker spending is

$$\min\{1, 4\varepsilon - W\} = \min\{1, 2\varepsilon\} = 2\varepsilon,$$

because $2\varepsilon < 1$. The residual active earning of s is

$$\min\{3, 2\varepsilon - W\} = 0.$$

The floor buyer has inverse MBB value $\lambda_F = p_r/u_{F,r} = 4\varepsilon/4 = \varepsilon$, while $p_s/u_{F,s} = 2\varepsilon > \varepsilon$, so only r is MBB for F . Its active budget is

$$\min\{2, 2\lambda_F\} = 2\varepsilon,$$

exactly the residual earning of r . Hence the shifted floor equations are satisfied at $\lambda_F = \varepsilon$, in addition to the intended large-scale solution near $\lambda_F = 1$. $\square$

Proposition 1 is the reason the informal sentence “give each blocker a tiny budget and absorb the slack” is not a proof. A successful strict-positive-budget reduction would need a further gadget that prevents these low-scale equilibria or otherwise proves that they cannot extend to a full market equilibrium. I did not obtain such a gadget here. The verified conclusion of this note is therefore exactly Theorem 1, plus the positive-budget accounting lemma and the obstruction above.

Acknowledgment of scope

The reduction is directly relevant to the future-direction question in Garg–Hoefer–Mehlhorn: it proves the NP-hard alternative for the nonnegative-budget convention. Because the formal model in the paper states $m_i > 0$, this note should be described as partial progress toward the paper’s strict-positive-budget question, not as a complete resolution of it.

References

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